

Continuum Mechanics HW4

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Technion

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1 Jet Incident onto a Plane

1.1 force on surface

The incident velocity is, expressed as a vector

$$\mathbf{v}_{in} = U (-\cos \theta, -\sin \theta) \quad (1)$$

and the outgoing velocities are

$$\mathbf{U}_1 = (-U_1, 0), \quad \mathbf{U}_2 = (U_2, 0) \quad (2)$$

from conservation of mass

$$Uh = U_1 h_1 + U_2 h_2 \quad (3)$$

so

$$h = h_1 + h_2 \quad (4)$$

using Bernoulli's theorem since the flow is steady and ideal,

$$p_0 + \frac{1}{2}\rho U^2 = p_0 + \frac{1}{2}\rho U_1^2 = p_0 + \frac{1}{2}\rho U_2^2 \quad (5)$$

so

$$U = U_1 = U_2 \quad (6)$$

the momentum flux going into the system is

$$\dot{\mathbf{P}}_{in} = \rho h U^2 (-\cos \theta, -\sin \theta) \quad (7)$$

and the momentum flux leaving the system is

$$\dot{\mathbf{P}}_{out} = \rho U^2 (h_2 - h_1, 0) \quad (8)$$

so the force the plate applies on the liquid is

$$\mathbf{F} = \dot{\mathbf{P}}_{out} - \dot{\mathbf{P}}_{in} = \rho U^2 (h_2 - h_1 + h \cos \theta, h \sin \theta) \quad (9)$$

so

$$\mathbf{f} = \rho U^2 ((h_1 - h_2 - h \cos \theta) \hat{\mathbf{x}} - h \sin \theta \hat{\mathbf{y}}) \quad (10)$$

Since the jet is of an ideal fluid, the only stress it can transmit to the plate is from pressure, which always acts along the normal. so,

$$\mathbf{f} = -\rho U^2 - h \sin \theta \hat{\mathbf{y}} \quad (11)$$

1.2 thickness as a function

we already know

$$h = h_1 + h_2 \quad (12)$$

and we can add

$$h_1 - h_2 = h \cos \theta \quad (13)$$

from the parallel part of the force being zero. We can now insert the above relations into each other and get

$$h - h_2 - h_2 = h \cos \theta \implies h_2 = \frac{h}{2} (1 - \cos \theta) \quad (14)$$

$$h_1 - h + h_1 = h \cos \theta \implies h_1 = \frac{h}{2} (1 + \cos \theta) \quad (15)$$

2 Boundary Layer on a Surface Attached to a Sink

2.1 Boundary conditions

The boundary conditions are non-penetration, no slip, and the condition to match the outer flow:

$$v_y(x > 0, y = 0) = 0 \quad (16)$$

$$v_x(x > 0, y = 0) = 0 \quad (17)$$

$$v_x(x, y) \rightarrow U(x) = \frac{m}{x} \text{ as } y/\delta(x) \rightarrow \infty \quad (18)$$

2.2 thickness

The second principle is that inside the layer the inertial term and the viscous terms are of the same order. We now look at the approximations

$$v_x \sim U, \quad \partial_x \sim 1/x, \quad \partial_y \sim 1/\delta \quad (19)$$

which give

$$v_x \frac{\partial v_x}{\partial x} \sim \frac{U^2}{x}, \quad \nu \frac{\partial^2 v_x}{\partial y^2} \sim \frac{\nu U}{\delta^2} \quad (20)$$

balancing them

$$\frac{U^2}{x} \sim \frac{\nu U}{\delta^2} \Rightarrow \delta \sim \sqrt{\frac{\nu x}{|U(x)|}} = \sqrt{\frac{\nu x}{|m|/x}} \Rightarrow \delta(x) = x \sqrt{\frac{\nu}{|m|}} \propto x \quad (21)$$

2.3 showing equation

The stream function from class is

$$\Psi(x, y) = -U(x)\delta(x)f(\eta) \quad (22)$$

but

$$U(x)\delta(x) = \frac{m}{x} x \sqrt{\frac{\nu}{|m|}} = \sqrt{\nu|m|} \quad (23)$$

which is constant, meaning Ψ is a function of η alone. translating the boundary conditions to conditions on f we get

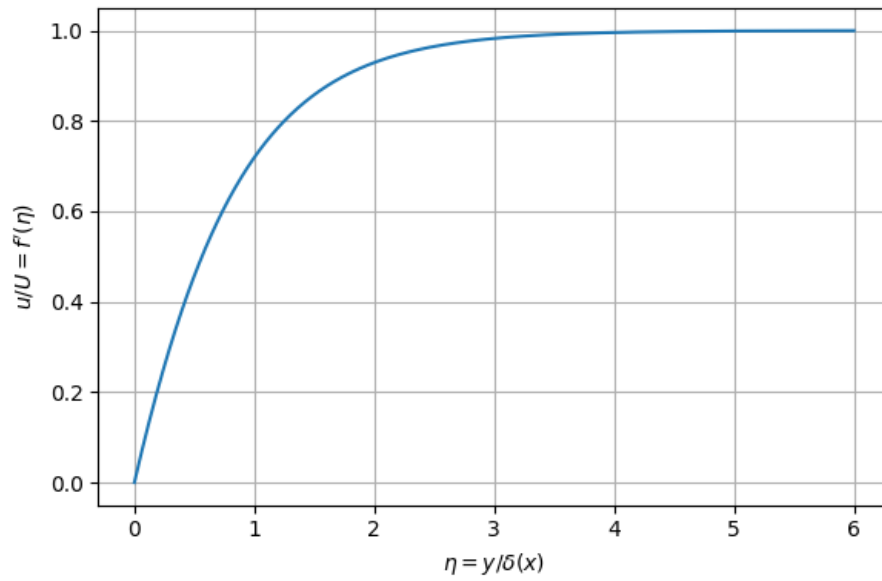
$$f(0) = 0, \quad f'(0) = 0, \quad f'(\infty) = 1 \quad (24)$$

substituting this into the first Prandtl equation

$$f''' - (f')^2 = -1 \quad (25)$$

2.4 numerical

the plot is



The velocity rises to be basically the same as the outside velocity at approximately $\eta = 3.5$. Since $\eta = \frac{y}{\delta(x)}$ it means we need to move away only a little more than δ and the width of the boundary layer in the y direction is proportional to δ .

3 Strain Tensor

3.1 small deformation

The deformation is

$$\mathbf{u} = (x', y', z') - (x, y, z) = \left(x \left(e^{-y/D} - 1 \right), y \left(e^{-x/D} - 1 \right), 0 \right) \quad (26)$$

for the deformation to be small, we want

$$\frac{\partial u_i}{\partial x_j} \ll 1 \quad (27)$$

and since the coordinates are of order of magnitude of L we want

$$\frac{L}{D} \ll 1 \quad (28)$$

or

$$L \ll D \quad (29)$$

3.2 nonlinear components

in the small deformation case we found that

$$L/D \ll 1 \quad (30)$$

and the nonlinear term goes like

$$\frac{\partial u_k}{\partial x'_i} \frac{\partial u_k}{\partial x'_j} \sim (L/D)^2 \ll L/D \quad (31)$$

so it's very negligible.

3.3 Linear Strain Tensor

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (32)$$

where

$$\frac{\partial u_x}{\partial x} \simeq -y/D \quad \frac{\partial u_x}{\partial y} \simeq -x/D \quad \frac{\partial u_y}{\partial x} \simeq -y/D \quad \frac{\partial u_y}{\partial y} \simeq -x/D \quad (33)$$

so

$$\varepsilon_{ij} = \begin{pmatrix} -y/D & -\frac{x+y}{2D} & 0 \\ -\frac{x+y}{2D} & -x/D & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (34)$$

3.4 volume change

we saw in the lecture that the change in volume of an element is

$$\frac{dV}{V} = \varepsilon_{ii} \quad (35)$$

and here

$$\varepsilon_{ii} = -y/D - x/D = -\frac{x+y}{D} \quad (36)$$

so the total change is

$$\Delta V = \int_0^L \int_0^L \int_0^L \left(-\frac{x+y}{D} \right) dx dy dz \quad (37)$$

which from symmetry (in the mathematical sense) we can rewrite as

$$\Delta V = \frac{-2}{D} \int_0^L \int_0^L \int_0^L x dx dy dz = -\frac{L^4}{D} \quad (38)$$

so the total volume is

$$V' = L^3 - \frac{L^4}{D} \quad (39)$$

4 Stress Tensor under Rotations

we can say

$$R_{\alpha_p} = \begin{pmatrix} \cos \alpha_p & -\sin \alpha_p \\ \sin \alpha_p & \cos \alpha_p \end{pmatrix} \quad (40)$$

so

$$\sigma'_{ij} = \begin{pmatrix} \cos \alpha_p & -\sin \alpha_p \\ \sin \alpha_p & \cos \alpha_p \end{pmatrix} \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{pmatrix} \begin{pmatrix} \cos \alpha_p & -\sin \alpha_p \\ \sin \alpha_p & \cos \alpha_p \end{pmatrix} \quad (41)$$

$$= \begin{pmatrix} \cos \alpha_p & -\sin \alpha_p \\ \sin \alpha_p & \cos \alpha_p \end{pmatrix} \begin{pmatrix} \cos \alpha_p \sigma_{xx} + \sin \alpha_p \sigma_{xy} & \cos \alpha_p \sigma_{xy} - \sin \alpha_p \sigma_{xx} \\ \cos \alpha_p \sigma_{xy} + \sin \alpha_p \sigma_{yy} & \cos \alpha_p \sigma_{yy} - \sin \alpha_p \sigma_{xy} \end{pmatrix} \quad (42)$$

$$= \begin{pmatrix} \sigma_{xx} \cos^2 \alpha_p + \sigma_{yy} \sin^2 \alpha_p + 2\sigma_{xy} \sin \alpha_p \cos \alpha_p & (\sigma_{yy} - \sigma_{xx}) \sin \alpha_p \cos \alpha_p + \sigma_{xy} (\cos^2 \alpha_p - \sin^2 \alpha_p) \\ (\sigma_{yy} - \sigma_{xx}) \sin \alpha_p \cos \alpha_p + \sigma_{xy} (\cos^2 \alpha_p - \sin^2 \alpha_p) & \sigma_{xx} \sin^2 \alpha_p + \sigma_{yy} \cos^2 \alpha_p - 2\sigma_{xy} \sin \alpha_p \cos \alpha_p \end{pmatrix} \quad (43)$$

we can use the identities

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha, \quad \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha \quad (44)$$

we can have

$$\sigma_{12} = -\frac{\sigma_{xx} - \sigma_{yy}}{2} \sin 2\alpha + \sigma_{xy} \cos 2\alpha \quad (45)$$

4.1 principal axes

we require

$$\sigma_{12} = -\frac{\sigma_{xx} - \sigma_{yy}}{2} \sin 2\alpha + \sigma_{xy} \cos 2\alpha = 0 \quad (46)$$

so

$$\sigma_{xy} \cos 2\alpha = \frac{\sigma_{xx} - \sigma_{yy}}{2} \sin 2\alpha \quad (47)$$

dividing by $\cos 2\alpha$

$$\sigma_{xy} = \frac{\sigma_{xx} - \sigma_{yy}}{2} \tan 2\alpha \quad (48)$$

rearranging

$$\tan 2\alpha = \frac{2\sigma_{xy}}{\sigma_{xx} - \sigma_{yy}} \quad (49)$$

4.2 principal stress

we'll find the eigenvalues of the stress tensor

$$\det \begin{pmatrix} \sigma_{xx} - \lambda & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} - \lambda \end{pmatrix} = 0 \quad (50)$$

or

$$(\sigma_{xx} - \lambda)(\sigma_{yy} - \lambda) - \sigma_{xy}^2 = 0 \quad (51)$$

expanding

$$\lambda^2 - (\sigma_{xx} + \sigma_{yy})\lambda + \sigma_{xx}\sigma_{yy} - \sigma_{xy}^2 = 0 \quad (52)$$

solving

$$\lambda_{1,2} = \frac{\sigma_{xx} + \sigma_{yy}}{2} \pm \frac{1}{2} \sqrt{(\sigma_{xx} + \sigma_{yy})^2 - 4(\sigma_{xx}\sigma_{yy} - \sigma_{xy}^2)} \quad (53)$$

simplifying inside the root

$$(\sigma_{xx} + \sigma_{yy})^2 - 4(\sigma_{xx}\sigma_{yy} - \sigma_{xy}^2) = (\sigma_{xx} - \sigma_{yy})^2 + 4\sigma_{xy}^2 \quad (54)$$

so

$$\sigma_{1,2} = \frac{\sigma_{xx} + \sigma_{yy}}{2} \pm \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \sigma_{xy}^2} \quad (55)$$

4.3 angle of max shear stress

we know

$$\sigma_{12} = -\frac{\sigma_{xx} - \sigma_{yy}}{2} \sin 2\alpha + \sigma_{xy} \cos 2\alpha \quad (56)$$

taking the derivative

$$\sigma'_{12} = -(\sigma_{xx} - \sigma_{yy}) \cos 2\alpha - 2\sigma_{xy} \sin 2\alpha \quad (57)$$

extrema when

$$2\sigma_{xy} \sin 2\alpha = -(\sigma_{xx} - \sigma_{yy}) \cos 2\alpha \quad (58)$$

rearranging

$$\tan 2\alpha = -\frac{\sigma_{xx} - \sigma_{yy}}{2\sigma_{xy}} \quad (59)$$

we notice that this angle is very similar to the principal axis angle, so we compare

$$\tan 2\alpha_s = -(\tan 2\alpha_p)^{-1} \quad (60)$$

which implies the angles are perpendicular, so

$$\alpha_s = \alpha_p + \frac{\pi}{2} \quad (61)$$

the maximum magnitude for the shear stress is

$$|\sigma'_{12}| = \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \sigma_{xy}^2} \quad (62)$$

but from the expression for the principal stresses

$$\sigma_1 - \sigma_2 = 2\sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \sigma_{xy}^2} \quad (63)$$

so

$$|\sigma'_{12}| = \frac{1}{2} (\sigma_1 - \sigma_2) \quad (64)$$

4.4 if shear always zero

since

$$\sigma'_{12} = -\frac{\sigma_{xx} - \sigma_{yy}}{2} \sin 2\alpha + \sigma_{xy} \cos 2\alpha \quad (65)$$

this means

$$\frac{\sigma_{xx} - \sigma_{yy}}{2} \sin 2\alpha = \sigma_{xy} \cos 2\alpha \quad (66)$$

but since this is true for all α then

$$\frac{\sigma_{xx} - \sigma_{yy}}{2} = 0, \quad \sigma_{xy} = 0 \quad (67)$$

therefore

$$\sigma_{xx} = \sigma_{yy}, \quad \sigma_{xy} = 0 \quad (68)$$

so the stress tensor is of the form

$$\sigma_{ij} = \sigma_{xx} \delta_{ij} \quad (69)$$

and with a simple change of notation

$$\sigma_{ij} = -p \delta_{ij} \quad (70)$$

5 Hooke's Law from Free Energy

5.1 deriving Hookes law

The differential of F is

$$dF = d\left(\frac{\lambda}{2}\varepsilon_{ii}^2 + \mu\varepsilon_{ij}^2\right) \quad (71)$$

$$= \lambda\varepsilon_{kk} d\varepsilon_{ll} + 2\mu\varepsilon_{ij} d\varepsilon_{ij} \quad (72)$$

$$= \lambda\varepsilon_{kk}\delta_{ij} d\varepsilon_{ij} + 2\mu\varepsilon_{ij} d\varepsilon_{ij} \quad (73)$$

$$= (\lambda\varepsilon_{kk}\delta_{ij} + 2\mu\varepsilon_{ij}) d\varepsilon_{ij} \quad (74)$$

but by definition

$$dF = \sigma_{ij} d\varepsilon_{ij} \quad (75)$$

so

$$\sigma_{ij} = \lambda\varepsilon_{kk}\delta_{ij} + 2\mu\varepsilon_{ij} \quad (76)$$

5.2 comparison

in class we saw

$$\sigma_{ij} = \frac{E}{1+\nu}\varepsilon_{ij} + \frac{E\nu}{(1+\nu)(1-2\nu)}\varepsilon_{kk}\delta_{ij} \quad (77)$$

so we see the coefficients are

$$\mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \quad (78)$$

6 Elastic Modulus Tensor in a Two-Dimensional Crystal

6.1 Symmetries in square lattice

In the training we saw that the tensor has the symmetries

$$C_{ijkl} = C_{jikl} = C_{ijlk} = C_{klij} \quad (79)$$

Which leads to there already being only 6 independent constants. As for the symmetries of the square lattice, it has symmetries for mirrorings in the x and y directions, as well as rotations of multiples of $\pi/2$. The mirroring symmetry implies the remaining independent elements are

$$C_{xxxx}, \quad C_{yyyy}, \quad C_{xxyy}, \quad C_{xyzy} \quad (80)$$

because under mirroring $\varepsilon_{xy} \rightarrow -\varepsilon_{xy}$ but $\varepsilon_{xx/yy} \rightarrow \varepsilon_{xx/yy}$ so mixed terms of $\varepsilon_{xx/yy}\varepsilon_{xy}$ have to vanish. If we rotate the lattice by $\pi/2$ it's effectively the same as renaming x to y and vice versa, meaning

$$C_{xxxx} = C_{yyyy} \quad (81)$$

and we're left with three elements

$$C_{xxxx}, \quad C_{xxyy}, \quad C_{xyzy} \quad (82)$$

6.2 Triangular Lattice

The triangular lattice also has symmetry for mirroring in x and y , but the rotational symmetry is for rotations of $\pi/3$. We can start from the 4 independent elements

$$C_{xxxx}, \quad C_{yyyy}, \quad C_{xxyy}, \quad C_{xyzy} \quad (83)$$

and require the symmetry for $\pi/3$. The strain tensor transforms like a second rank tensor, so

$$\varepsilon'_{xx} = \varepsilon_{xx} \cos^2 \alpha + \varepsilon_{yy} \sin^2 \alpha + 2\varepsilon_{xy} \sin \alpha \cos \alpha \quad (84)$$

$$\varepsilon'_{yy} = \varepsilon_{xx} \sin^2 \alpha + \varepsilon_{yy} \cos^2 \alpha - 2\varepsilon_{xy} \sin \alpha \cos \alpha \quad (85)$$

$$\varepsilon'_{xy} = (\varepsilon_{yy} - \varepsilon_{xx}) \sin \alpha \cos \alpha + \varepsilon_{xy} (\cos^2 \alpha - \sin^2 \alpha) \quad (86)$$

denoting for convenience

$$A = C_{xxxx}, \quad B = C_{yyyy}, \quad C = C_{xxyy}, \quad D = C_{xyzy} \quad (87)$$

and using the free energy

$$F = \frac{1}{2}A\varepsilon_{xx}^2 + B\varepsilon_{xx}\varepsilon_{yy} + 2C\varepsilon_{xy}^2 + \frac{1}{2}D\varepsilon_{yy}^2 \quad (88)$$

and

$$\cos \pi/3 = 1/2, \quad \sin \pi/3 = \frac{\sqrt{3}}{2} \quad (89)$$

so

$$\varepsilon'_{xx} = \frac{1}{4}\varepsilon_{xx} + \frac{3}{4}\varepsilon_{yy} + \frac{\sqrt{3}}{2}\varepsilon_{xy} \quad (90)$$

$$\varepsilon'_{yy} = \frac{3}{4}\varepsilon_{xx} + \frac{1}{4}\varepsilon_{yy} - \frac{\sqrt{3}}{2}\varepsilon_{xy} \quad (91)$$

$$\varepsilon'_{xy} = \frac{\sqrt{3}}{4}(\varepsilon_{yy} - \varepsilon_{xx}) - \frac{1}{2}\varepsilon_{xy} \quad (92)$$

inserting into the free energy

$$F = \frac{1}{2}A \left(\frac{1}{4}\varepsilon_{xx} + \frac{3}{4}\varepsilon_{yy} + \frac{\sqrt{3}}{2}\varepsilon_{xy} \right)^2 + B \left(\frac{1}{4}\varepsilon_{xx} + \frac{3}{4}\varepsilon_{yy} + \frac{\sqrt{3}}{2}\varepsilon_{xy} \right) \left(\frac{3}{4}\varepsilon_{xx} + \frac{1}{4}\varepsilon_{yy} - \frac{\sqrt{3}}{2}\varepsilon_{xy} \right) \quad (93)$$

$$+ 2C \left(\frac{\sqrt{3}}{4}(\varepsilon_{yy} - \varepsilon_{xx}) - \frac{1}{2}\varepsilon_{xy} \right)^2 + \frac{1}{2}D \left(\frac{3}{4}\varepsilon_{xx} + \frac{1}{4}\varepsilon_{yy} - \frac{\sqrt{3}}{2}\varepsilon_{xy} \right)^2 \quad (94)$$

and we demand it is equal to the original free energy. rearranging

$$F' = \left(\frac{A}{32} + \frac{3B}{16} + \frac{3C}{8} + \frac{9D}{32} \right) \varepsilon_{xx}^2 \quad (95)$$

$$+ \left(\frac{3A}{16} + \frac{5B}{8} - \frac{3C}{4} + \frac{3D}{16} \right) \varepsilon_{xx}\varepsilon_{yy} \quad (96)$$

$$+ \left(\frac{\sqrt{3}A}{8} + \frac{\sqrt{3}B}{4} + \frac{\sqrt{3}C}{2} - \frac{3\sqrt{3}D}{8} \right) \varepsilon_{xx}\varepsilon_{xy} \quad (97)$$

$$+ \left(\frac{9A}{32} + \frac{3B}{16} + \frac{3C}{8} + \frac{D}{32} \right) \varepsilon_{yy}^2 \quad (98)$$

$$+ \left(\frac{3\sqrt{3}A}{8} - \frac{\sqrt{3}B}{4} - \frac{\sqrt{3}C}{2} - \frac{\sqrt{3}D}{8} \right) \varepsilon_{yy}\varepsilon_{xy} \quad (99)$$

$$+ \left(\frac{3A}{8} - \frac{3B}{4} + \frac{C}{2} + \frac{3D}{8} \right) \varepsilon_{xy}^2 \quad (100)$$

we get the equations

$$\frac{A}{32} + \frac{3B}{16} + \frac{3C}{8} + \frac{9D}{32} = \frac{A}{2} \quad (101)$$

$$\frac{3A}{16} + \frac{5B}{8} - \frac{3C}{4} + \frac{3D}{16} = B \quad (102)$$

$$\frac{\sqrt{3}A}{8} + \frac{\sqrt{3}B}{4} + \frac{\sqrt{3}C}{2} - \frac{3\sqrt{3}D}{8} = 0 \quad (103)$$

$$\frac{9A}{32} + \frac{3B}{16} + \frac{3C}{8} + \frac{D}{32} = \frac{D}{2} \quad (104)$$

$$\frac{3\sqrt{3}A}{8} - \frac{\sqrt{3}B}{4} - \frac{\sqrt{3}C}{2} - \frac{\sqrt{3}D}{8} = 0 \quad (105)$$

$$\frac{3A}{8} - \frac{3B}{4} + \frac{C}{2} + \frac{3D}{8} = 2C \quad (106)$$

focusing on two equations

$$A + 2B + 4C - 3D = 0 \quad (107)$$

$$3A - 2B - 4C - D = 0 \quad (108)$$

adding them together gives

$$A = D \tag{109}$$

inserting this back

$$-2A + 2B + 4C = 0 \tag{110}$$

rearranging

$$C = \frac{A - B}{2} \tag{111}$$

so we're left with two independent components.

6.3 isotropic material

in an isotropic material we have symmetries for rotation for every rotation. I stop here because I'm short on time, unfortunately.